nPnB Graph Explorer
Quick User Guide

1 Graph Exploration at Different Description Scales

Networks are one of the main conceptual structures used to model complex systems, where nodes represent the lowest-level entities of the system and edges represent their interactions. Graph clustering algorithms are then used to identify modules: over-densely connected sets of nodes representing higher-level entities of the system.

bec is a C++ graph clustering program that identifies Entities in a network through the concept of **Description Scales** within the **nPnB framework**.

> **nPnB:** B. Gaume, I. Achitouv, and D. Chavalarias: *Two antagonistic objectives for one multi-scale graph clustering framework.* Nature Scientific Reports, 15 (2025), p. 13368. `https://hal.science/hal-05046050v1`
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> **bec:** B. Gaume: *Finding optimal clusterings in the nPnB framework is hard. BEC.2: an algorithm to find high-performing ones.* Forthcoming (2026).
>
> **Description Scales:** B. Gaume: *Defining the appropriate scales for describing a complex system.* Forthcoming (2026).

To visualize graphs at different Description Scales, we use the 3D Force-Directed Graph Layout developed by Vasco Asturiano: `https://github.com/vasturiano`

• Two nodes sharing the same color belong to the same **Entity** at the active **Description Scale** s. The **[FrozCol]** button allows you to freeze node colors at the active Description Scale s.

• The **[ScaleLayout]** button allows you to weaken the attractive force of links whose endpoints do not belong to the same Entity at the active Description Scale. This modifies the **Graph Layout** according to the active Description Scale s.

• Combining **[FrozCol]** at a Description Scale s_1 with **[ScaleLayout]** at another Description Scale s_2 allows simultaneous observation of the graph at two different Description Scales. This often reveals the fractal organization present in many real-world graphs.

Important
Node actions are centered on a selected node. Background actions apply to the graph globally.

2 Mouse Actions

2.1 Description Scale

You can activate a Description Scale by clicking on it. The corresponding clustering score is displayed below: on the left for partition clustering, and on the right for overlapping clustering.

2.2 Node

Left Click	Zoom on the selected node.
Right Click	Open a visibility panel centered on the selected node. Typical operations include showing or hiding neighbours, and showing or hiding members of the node module at the selected Description Scale.
Ctrl + Left Click	Open the node properties panel. This panel allows updating node visibility, label visibility, and large label visibility.
Ctrl + Right Click	Toggle label visibility for the selected node.

2.3 Background

Left Click	No action. Useful for clicking on empty space.
Double Left Click	Toggle graph rotation.
Right Click	Open the global visibility panel. Typical operations include node degree filtering, label regular-expression filtering, and graph-wide visibility criteria.
Shift + Left Click	Place all visible nodes in the center of the scene.
Ctrl + Left Click	Hide labels on all currently visible nodes.
Ctrl + Right Click	Show labels on all currently visible nodes.

3 Toolbar

CONTROL	Open the control panel.
Node Drag	Toggle node dragging.
Rotation	Toggle graph rotation (+ Increase rotation speed, - Decrease rotation speed)
ScaleLayout	Modify the Graph Layout according to the active Description Scale. Links connecting nodes belonging to different Entities are progressively weakened. (+ weakens more, - weakens less)
FrozCol	Freeze node colors at the active Description Scale.
Export	Export the current graph view as an image.

4 Tips

- Use node actions for local exploration.
- Use background actions for global operations.
- Keep labels hidden when exploring large graphs.
- Display labels only when additional details are required.
- Compare graph organizations across different Description Scales.